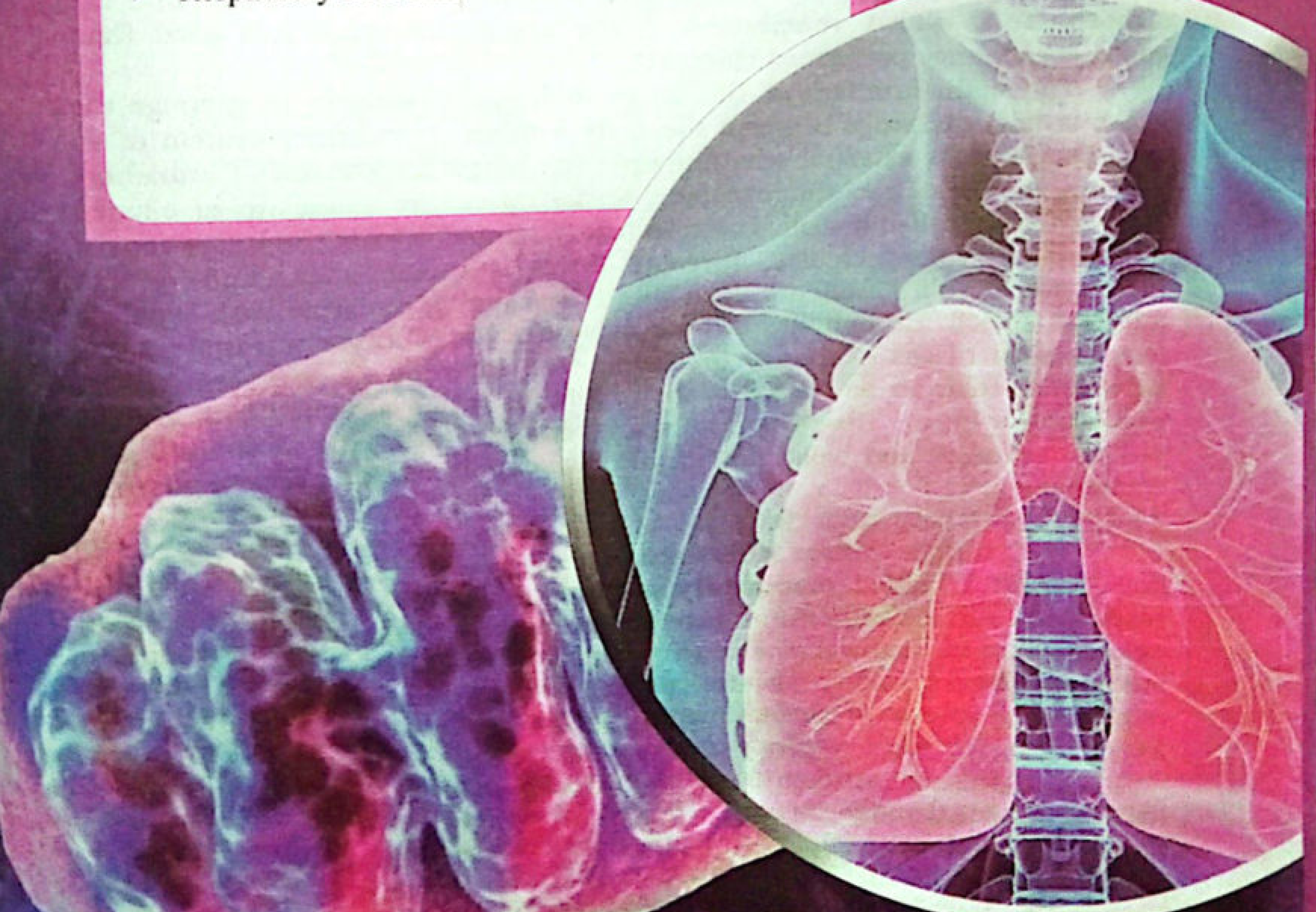
GASEOUS EXCHANGE

Chapter 14

Major Concept

In this Unit you will learn:

- Respiration and Respiratory System of Man
- Mechanism of Transport of Gases
- Respiratory Disorders



14.1 RESPIRATION

Respiration is a very important process in which the cell or cells of an organism (s) gain energy (ATP) rich molecules by oxidizing food molecules. In human being the food molecule is chiefly used for this purpose. During aerobic respiration glucose in our cells oxidized with the help of oxygen, the energy is released along with by-products carbon dioxide and water. Therefore, the many of cells in an organism body require abundant amount and continuous supply of oxygen (O_2) to perform the bodily functions. We can't live without oxygen for more than a few minutes.

14.1.1 Ventilation and Gaseous exchange

As you already learnt that we provide glucose to our cell through nutrition process. However, in order to supply oxygen to our cells, we have to bring air in to our lungs through the air passage ways. Thus, the movement of air from the atmosphere into the lungs and back into the atmosphere is called **ventilation**. If the ventilation quality is good than gaseous exchange performed efficiently.

The main function of the lungs in human being is to manage the gaseous (air) exchange between the body internal circulatory system of an organism and its external environment. The lungs are consist of branching airways that end into microscopic sac like structure made up of ciliated epithelial cells called alveoli. The respiratory related area of the lungs consist on respiratory bronchioles, alveolar ducts, alveolar sacs, and a thin membrane alveoli. Alveoli are lined with a thin film of moisture with separate adjacent alveoli. Bronchioles of both sides and large airways transvers gas (air) to sites of gas exchange in alveoli. The interchange of gases (air) in between alveolar air and blood of the capillaries which attached with alveoli, occur in the lungs. The alveoli must be ventilated (the flow of air into and out of the lungs) and blood flow in alveolar capillaries for effective gaseous exchange.

14.1.2 Respiratory surface:

The surface where the exchange of gases take place with its environment is called **respiratory surface**. Each organism must need a respiratory surface to perform gaseous exchange with its environment. For example skin in earthworm, alveolar membrane etc. Our respiratory surface is alveoli. The respiratory surface should be moist, thin, permeable and large in relation to volume of the organism.

14.1.3 Characters of respiratory surface:

Moist: Respiratory surface (alveoli) must be moist which allow to pass dissolved gases through them.

Permeable: The permeability of respiratory surface should be permeable to gases (O_2, CO_2etc).

Thin: Being thin the respiratory surface reduce the diffusion distance.

Large surface area: The respiratory surface must be large in size in accordance to the body size of an organism. In our lungs we have millions of alveoli whose collective surface area is several folds larger in relation to our total volume.

In case of higher animal like human, the respiratory surface is facilitated by following:

Significant blood supply: The another character of respiratory surface is to initiating highly concentrated oxygenated blood is taken away from the lungs and carbon dioxide rich blood is taken to the lungs.

A large diffusion gradient: For the purpose of continuous breathing, oxygen concentration in the alveoli is higher than in the capillaries, so oxygen moves from the alveoli to the blood. Carbon dioxide diffuses in the opposite direction (Fig:14.1).

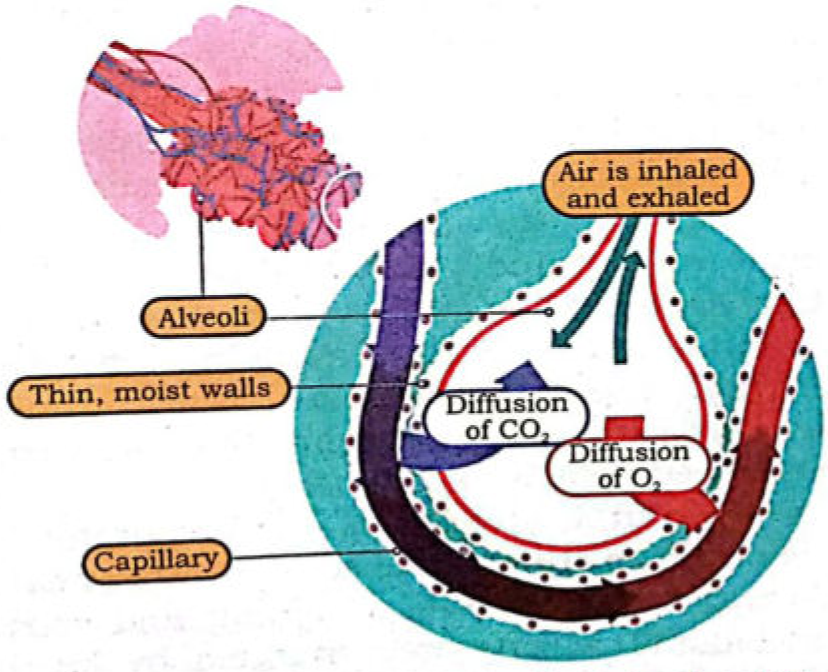


Fig: 14.1 Respiratory surface: An alveolus

14.1.4 Main Components of Human respiratory system

The human respiratory system consists of a complex set of organs and tissues. This system carries air from the environment to the lungs and vice versa. The human lungs are the paired, soft, spongy and highly vascularized organ in structure, located inside thorax (chest). Lungs are the primary organ of the respiratory system. However, the external nostril **(nose)**, **pharynx**, **larynx**, and **trachea** are the other respiratory accessory organs.

The human respiratory system is anatomically differentiated into upper and lower respiratory air passages. The upper airways are consist of nose, nasal cavity, and Pharynx while the lower air passage are the larynx, bronchial branches and the pair of lungs. The respiratory system starts from the nose. During respiration, air enters into the nose through the paired opening called **external nostrils** or nares. The nose is the only external part of our respiratory system. Interiorly nose consists of the nasal cavity, a bony partition that separates the nasal cavity into nasal chambers by a midline cartilaginous **nasal septum**. The hairs of nasal cavity and its ciliated cells trap the dust particle and decontaminate the gas (air).The nasal chambers run forward through internal nares into a tube like cavity at behind of the mouth. This cavity is called **pharynx**. The muscular tube pharynx is long about 13 cm, which serves as a common pathway for both the respiratory and digestive system. The external air enters from the nasal cavity to the pharynx. Then air move from pharynx to the upper portion, **nasopharynx** and **larynx** from **laryngopharynx**. Normally air enters the pharynx through the nose, but it may also enter by the mouth if the nasal sections are blocked. **Larynx** or **voice box**, is the component of lower respiratory track located at the entrance of the trachea. The flap like pairs of tissues called **vocal cords,** present in the larynx, which produce sounds when air expelled from it. From the pharynx, air enter into the **trachea** or windpipe. Trachea is the main airway to the lungs, it is a fibro-cartilaginous tube about 10-11 cm long. Trachea has about 16 to 20 "C" shaped cartilaginous ring, provide rigidity and prevent trachea collapsing. Before entering into lungs, trachea bifurcate into two bronchi. A singular right bronchus is broader, shorter and straighter than the left one bronchus. Each bronchi branches into smaller **bronchioles.** These bronchioles form the bronchial tree which terminates at the **alveoli**, the microscopic air sacs where gases are exchanged between air and blood.

The 12 pairs of ribs form the rib cage. It provides protection to lungs and heart. The wall of thoracic cavity is attached with **inter-costal muscles** which are associated with rib cage, vertebral column and sternum (chest bone). Thoracic and abdominal regions are separated by muscular sheet called **diaphragm**. The pleural membrane in thoracic region called **visceral pleura** wraps around each lung and separates from the thoracic wall. The other membrane is **parietal pleura,** lies the inside of the chest wall. The small space between the visceral and parietal pleurae is the **pleural cavity,** filled with serous fluid which serves as a lubricant to the **lungs** and allows optimal expansion and contraction during breathing mechanism.

14.1.5 Lungs

Lungs are the main respiratory organs. The left and right lungs are somewhat dissimilar in structure. The left lung is smaller than the right lung, because left lung is slightly displaced by the heart which occupies more space on the left side in thorax. Each lung consist on smaller units, called **lobes.** Three lobes are in right lung, called **superior, middle,** and

inferior lobes. Two lobes are found in left lung **called, superior** and **inferior lobes**. Each lobe is supplied by the secondary bronchi.

(i) Right lung

The three main lobes of right lung (fig: 14.2).

1.Superior lobe
2. Middle lobe
3. Inferior lobe

(ii) Left lung

The two main lobes of left lung (fig: 14.2).

1.Superior lobe
2. Inferior lobe

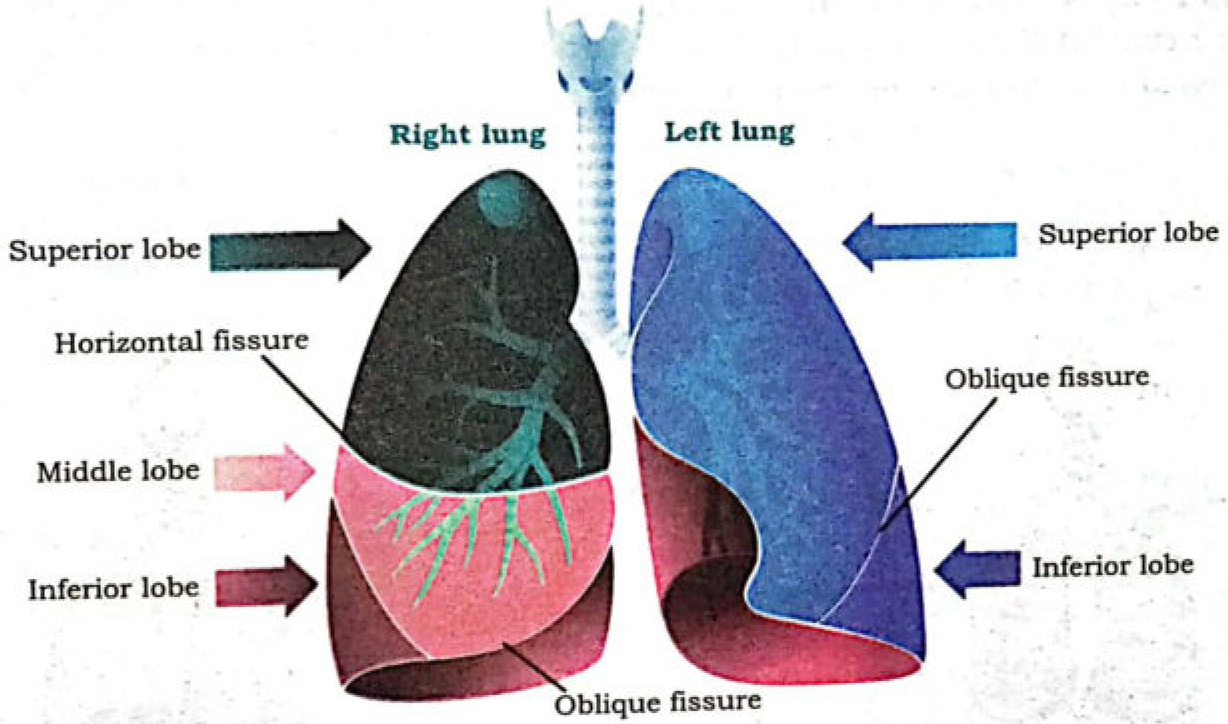


Fig: 14.2 Different lobes in both lungs

14.1.6.3 The major Functions of the respiratory system:

The respiratory system performs the following major functions:

Oxygen supplier:
It provides a continuous supply of oxygen to all tissues.

Withdrawal:
Removal of by-product, carbon dioxide (CO_2).

Conversation of Gas:
The mechanism of gas exchange between the internal and external environment of the body is regulate through respiratory surfaces.

Humidifier:
The respiratory system performs as a humidifier. It has the capability to humidify and keep the air warm which inhale from the external environment.

14.1.7 Ventilation (Breathing)

Ventilation is the process in which air (gases) goes in and out of the lungs. During this process, various respiratory organs are involved. This process goes on continuously throughout the life of an organism. The mechanism of movement of air from outside to deep inside and vice versa.

Ventilation (breathing) consist of two phases:

1. **Inhalation:** The taking in air from the atmosphere up to the alveoli.
2. **Exhalation:** Giving out of air from alveoli to the atmosphere.

Alive person breathes continuously. One breathing cycle comprises one complete inhalation and exhalation. The person breathes how much in every minute it's called breathing rate. Breathing rate varies upon a person's activity. It rises during running, walking or after a heavy exercise. The adult normal breathing rate is about 12-20 times per minutes and it may exceed or decrease due to different perspectives.

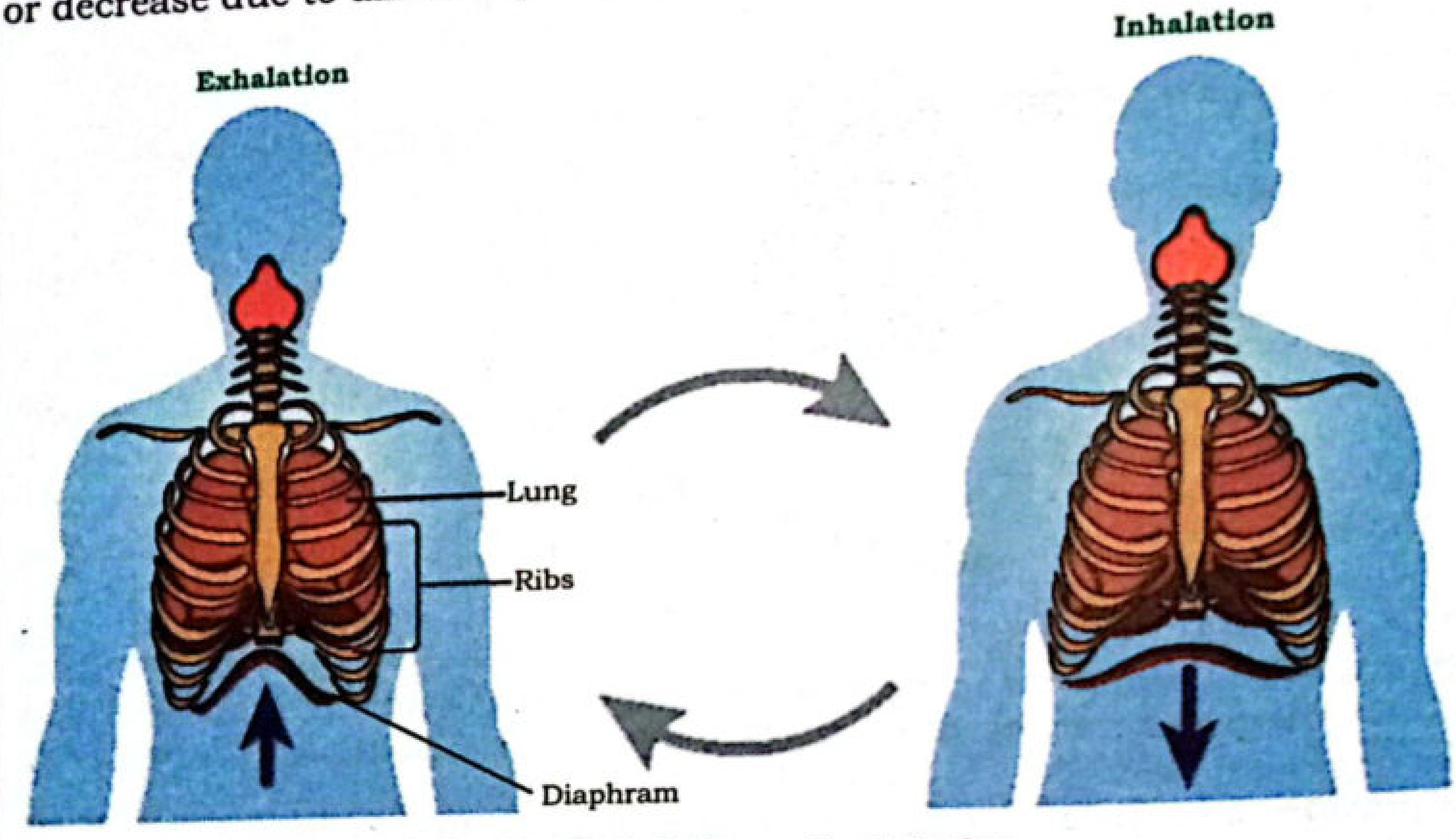


Fig: 14.3 inhalation and exhalation

14.1.7.1Mechanism of ventilation (Breathing)

The pressure of lungs change when we breathe in and out. So essentially in the **inspiration**, there is a fall in gaseous tension of the alveolar sac which falls when the air launches into the lungs. In **expiration**, the pressure of alveoli is exceed than atmospheric pressure, the air is goes out from the lungs. This type of breathing mechanism is called **negative pressure breathing** (which is the decrease pressure in the region of thoracic cavity in relation to external atmospheric pressure).

14.1.7.2Inspiration (Inhalation)

Basically, the inspiration is the active process (energy consuming mechanism). During this process, the thoracic cavity is increased in its size (volume) due to the contraction of **external inter–costal muscles**, diaphragm and relaxation of **internal inter-costal muscles**. Contraction of external inter–costal muscles moves the ribs outward and as well as sternum also progressively move in upward direction while shrinkage of the diaphragm makes it levelled. Subsequently, the capacity is increased in thoracic cavity, so negative pressure is developed inside the thoracic and eventually in the lungs. The expansion of the lungs decreases the air pressure inside the lungs. So ultimately air rushes into the lungs up to the alveolar sac through the respiratory passage where gaseous exchange take place (Fig. 14.3).

14.1.7.3Expiration(Exhalation)

It is the passive phase of respiration. In this process gaseous exchange completed in the lungs and air is thrown out which is just reversed of inspiration. Expiration takes place due to the decreased volume of the thoracic cavity. It is due to the relaxation of external inter-costal and the contraction of internal inter-costal muscles, which move ribs and sternum inward and downward respectively. Comparably, diaphragm also relaxes which makes it dome shaped, thus the volume of the thoracic cavity is reduced. As an outcome, lungs are pressed so the air along with water vapours is expire outside through respiratory pathway (Fig. 14.4).

14.1.7.4 Lung volume and capacity

Respiratory volumes are also known as lung volumes. In Human adult, lung capacity is six liters of air. Measurement of lung volume is an essential part to observe pulmonary function. These volumes vary due to some factors: race, gender, age, body composition and respiratory diseases etc. Lung volume measures by a special instrument, **spirometry**. The lung capacities of human can be describe by the following manner.

Total lung volume of air is about 5000 milliliters (6 liter), which generally enters in the lungs during breathing. In normal breathing, human takes in and gives out air approximately 450 to 500 milliliters. This volume is called **Tidal volume**. On other hand, **vital volume** is the maximum air inspired and expired during deep breath. This volume is regarding 5000 milliliters (ml). **Expiratory reserve volume** is the valuable quantity of air which energetically respire up to 1200-1500 milliliters. **Inspiratory reserve volume** is the extra air volume which is 2000 milliliters. **Residual volume** that remains in the lungs is about 1000 milliliters and cannot allow the thorax to collapse. The amount of residual volume can be changed due to the age and diseases. All these respiratory volumes can be describe according to the person's respiratory strength. (Fig: 14.4)

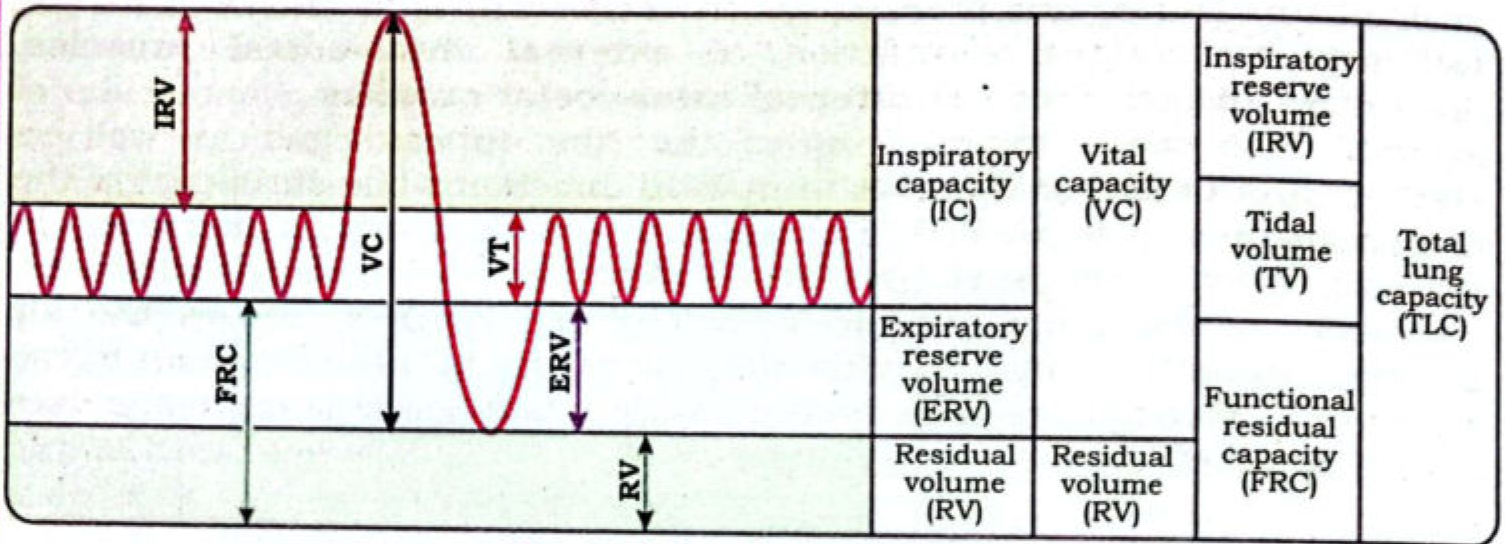


Fig: 14.4 Respiratory volumes

14.1.7.5Controlling mechanism of Breathing

Our breathing process can control for a very minimal duration or we can respire slower, faster and deeper at our will. This phase is called **voluntary control**. But generally our normal breathing is controlled automatically. This phase of breathing control is called **involuntary control**. The automatic breathing is maintained by the mutual association of respiratory and cardiovascular systems. It is observed that high concentration of carbon dioxide (CO_2) and H^+ in blood are stimuli to increase the rate of breathing. The accumulation of carbon dioxide (CO_2) and H^+ are monitored by specific chemoreceptors called **aortic** and **carotid bodies**. These bodies are found in aorta and carotid arteries, respectively. The lower part of brain medulla oblongata is responsible to detect any change in carbon dioxide and H^+ concentration in cerebro-spinal fluid. So this impulse (messages) sent to inter-costal muscles and diaphragm to increase the breathing rate, accordingly.

14.2 TRANSPORTATION OF GASES IN HUMAN

The exchange of Gases is the one of the vital process of respiration. Transportation of gases in all living creatures is a well-organized process. Oxygen (O_2) and carbon dioxide (CO_2) are the sources that transported in the human body by blood circulation and exchanged in the alveolar membrane by simple diffusion.

14.2.1 Transport of Oxygen

The inhaled air into lungs air passage has high concentration of oxygen which produces the concentration difference across the respiratory surfaces. So oxygen rushes into blood capillaries around the alveoli. Now the blood is oxygenated and converted into bright red colour. This oxygenated blood is then transported to body cells. Around 97 percent oxygen is disperse by red blood cells and the remaining 3 percent oxygen gets dissolved and transported through blood plasma. Oxygen (O_2) molecules attached to haemoglobin molecules of RBCs and form compound, oxyhaemoglobin. Oxyhaemoglobin delivers oxygen molecule to all body cells for cellular respiration. At the level of cells, oxygen molecules detached from haemoglobin and diffuse into the cells. The diffused oxygen breaks down the glucose molecules to release CO_2, water and energy. The body utilizes the energy while tissues diffuse the carbon dioxide.

$$\underset{\text{Haemoglobin}}{Hb} + \underset{\text{Tissues}}{\overset{\text{Lungs}}{4O_2}} \rightleftharpoons \underset{\text{Oxyhaemoglobin}}{Hb(O_2)_4}$$

14.2.2 Transport of Carbon dioxide

Due to the high concentration of CO_2 is diffuses out from the tissues into the blood as a by-product. Now blood is deoxygenated and collected from all part of the body cells and returned to the lungs. The color of deoxygenated blood is dark maroon. There are three ways to transport carbon dioxide into bloodstream:

1. As bicarbonate
2. As carbaminohemoglobin (bound to hemoglobin or other proteins),
3. As dissolved gas.

About 70% of carbon dioxide is transported by RBCs water in the presence of **carbonic anhydrase** enzyme. This enzyme reacts with carbon dioxide and form carbonic acid (H_2CO_3). Later, carbonic acid dissociates into hydrogen (H^+) and bicarbonate ($HCO3^-$) ions. Bicarbonate ions combine with sodium and potassium ions to form sodium bicarbonate and potassium bicarbonate respectively.

$$CO_2 + H_2O \xrightarrow{\text{Carbonic anhydrase}} H_2CO_3 \rightarrow HCO^-_3 + H^+$$
$$HCO_3^- + K^+ \rightleftharpoons KHCO_3$$
$$HCO_3^- + Na^+ \rightleftharpoons NaHCO_3$$

Approximately, 20 percent carbon dioxide combines with hemoglobin in the red blood cell to form carbaminohemoglobin. This compound is dissociated into CO_2 and Hb and CO_2 move into alveoli which is later brought in the lungs where its concentration is lower but partial pressure of oxygen is high in alveoli. So dissociated carbon dioxide is exhaled.

$$CO_2 + Hb \rightleftharpoons HbCO_2$$

The remaining 10% carbon dioxide dissolves in the plasma water of the blood and then transported to the lungs.

14.2.3 Role of respiratory pigments

A respiratory pigment is a molecule which increases the oxygen carrying capacity of blood and tissues for example haemoglobin and myoglobin. All of these are iron bounded proteins which combine readily with oxygen.

Haemoglobin

Haemoglobin (Hb) is the respiratory pigment present in RBCs of all vertebrate including human beings. Haemoglobin is the conjugated protein molecule. Two pairs of polypeptide chains are constituted Haemoglobin. Each chain bears an iron-containing heme group. Each hemoglobin molecule is•able to transport four oxygen molecules. That carries oxygen from the lungs to the body's tissues by forming a loose temporary compound called oxyhaemoglobin. This temporary compound is dissociates releasing oxygen, which enters in our tissues. Haemoglobin (Hb)brings back carbon dioxide from the tissues back to the lungs (fig: 14.5)

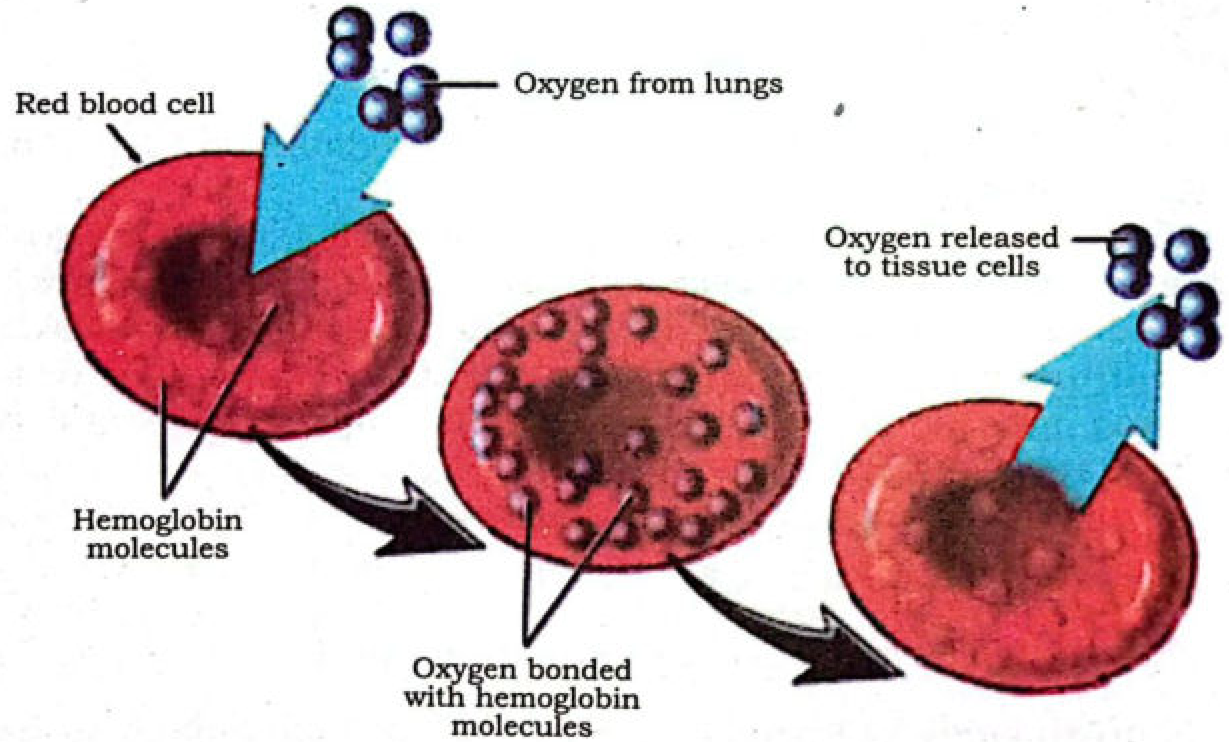


Fig: 14.5 Formation and dissociation of oxyhaemoglobin

14.2.3.2 Myoglobin

Myoglobin is a single-chain globular protein and smaller than haemoglobin. It is found in muscles and binds with oxygen more tightly than haemoglobin. It also consists of a heme group (an iron-containing organic compound). One hemebinds one molecule of O_2.

14.3 SINUSES

Sinuses are hollow air-filled cavities. These hollow cavities found in skull and linked to the nasal air passage. Human has four pairs of nasal cavities (Fig: 14.6)

1. **Frontal sinus**, in the forehead region
2. **Maxillary sinus**, in the behind cheeks
3. **Ethmoid sinus**, between the eyes
4. **Sphenoid sinus**, located in deep behind the ethmoids

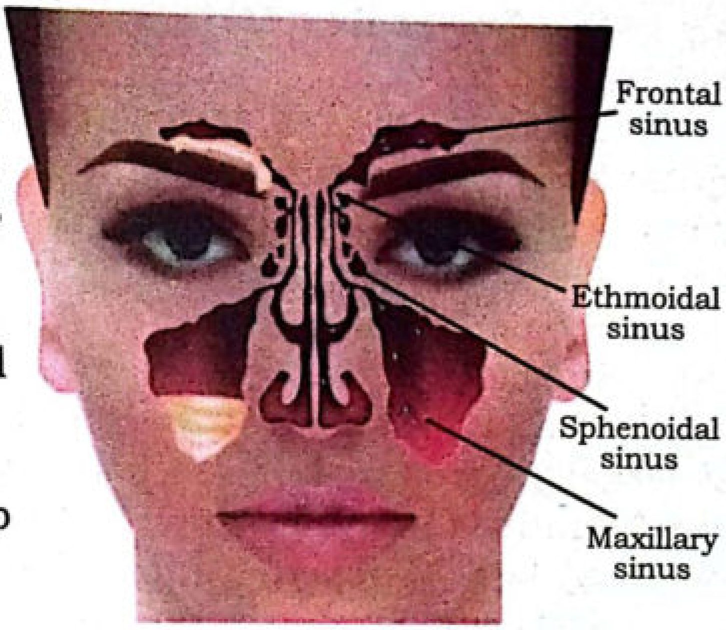


Fig: 14.6 Nasal cavities

14.3 RESPIRATORY DISORDERS

The respiratory disorder are broadly categorized into two groups on the basis of location of respiratory tract.

14.4.1 Upper Respiratory Tract Infections

(i) Sinusitis

Sinusitis is the inflammation of sinuses, which are filled with air normally, but in the sinusitis condition these sinuses are filled with fluid due to this it may harbor pathogens.

Following reasons that can cause sinus blockage include:

- Common cold
- Allergic rhinitis, swelling in the lining of the nose.
- Nasal polyps, nose lining show small growths.
- Nasal septum physical deviation of nose, involving a dislocation of the nasal septum.

The main symptoms of sinusitis are:

Headache, fever, congestion (nasal stuffiness), cough, tooth pain, ear pain, eye pain and fatigue.

Use steam and saline nasal spray to wash nasal passage and consult your physician.

Otitis media

Otitis media is inflammation or infection in the middle ear. Cold, sore throat, or upper respiratory infection can cause this disease. This infection usually a result of a failure of the **eustachian tube**. This tube bridges the middle ear with the throat area and balance the pressure between the outer and the middle ear.

Fever, unbalancing, hearing problems, unusual irritability and ear painare the very common signs of otitis media.

The treatment of otitis media is depend on its type. Commonly antibiotic medicines by ear drops can be suggested by consultant.

(ii) 14.4.2 Lower Respiratory Infections

A(i) Pneumonia

Pneumonia is a lower respiratory infection due to the number of virus, bacteria and fungi. In this infection air sacs filled with secretions and other fluid (Fig: 14.7).

- Lobar pneumonia occur in the lobes of lungs.
- Bronchial pneumonia (bronchopneumonia) patches appear on both lungs.

The symptoms of bacterial pneumonia are headache, lips and fingernails become bluish in color, fever with cough and yellowish green or bloody mucus may produce with cough, confused mental state, heavy sweating, Loss of appetite, rapid breathing and shortness of breath.

Viral and bacterial pneumonia have same symptoms in early stage which may causes shortness of breath, headache, muscle pain and weakness.

In Mycoplasma pneumonia symptoms are observed acute cough with mucus.

In most cases of bacterial pneumonia, oral antibiotics can use but antibiotic medications don't work on viruses. Patient should hot drinks. refreshments. Humidification and steamy baths also very helpful to open airways blockage and easy respiring.

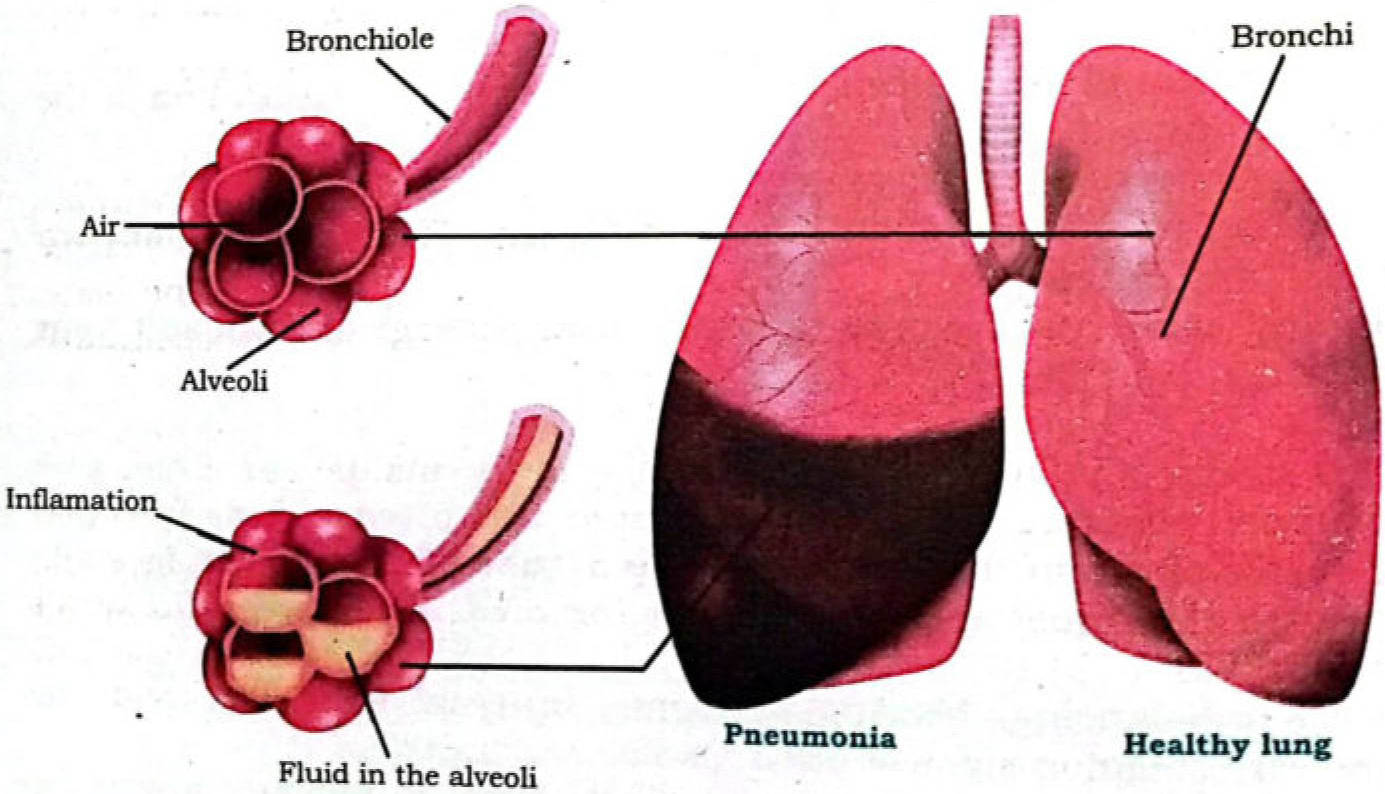


Fig: 14.7 Pneumonia

(ii) Pulmonary tuberculosis

Pulmonary tuberculosis (TB) is a bacterial chronic infection of the lungs and caused by *Mycobacterium tuberculosis.* The treatment of Pulmonary TB is possible but late treatment or untreated patient may get life threatening situation.

Symptoms including coughing up mucus (phlegm), coughing up blood, low-grade fever, chest pain and weight loss.

Pulmonary TB is curable with an early diagnosis and antibiotic treatment.

(iii) Emphysema

Emphysema is a condition of lung disease generally develops after many years of smoking. Emphysema is one of disease which belongs to group of chronic obstructive pulmonary disease (COPD).

In this condition wall of the air sacs become damaged. So, the alveoli cannot support the bronchial tubes. Due to the blockage of tubes too much air traps inside the lungs. There are fewer alveoli, less oxygen transport into bloodstream. It is irreversible condition.

Patient have no symptoms at the beginning but some specific symptoms can be observed in later stages like shortness of breath especially during physical activity, tightness in chest, frequent coughing or wheezing and mucus produces with cough. Some people with emphysema has respiratory infections such as cold and flu. In severe condition patient complains weakness in lower body muscles and weight loss.

Bronchodilators medicinal treatment recover the patient and relief from the coughing and short breathing. Antibiotics may be used for the remedy and steroid type drugs can also be inhaled like aerosol sprays which reduce swilling and may help in easy breath.

(iv) Lung cancer

It is abnormal proliferation of affected tissues in the lungs. The main reason of lung cancer is Smoking. About 80% of **lung cancer** caused death. Symptoms are coughing with blood, chest infections that keep coming back, pain when breathing or coughing, persistent breathlessness and continuous tiredness.

In lung cancer chemotherapy, radiation therapy, targeted therapy and surgery are acquires according to consultant.

(v) Smoking is dangerous for respiratory system

The effect of tobacco smoking on respiratory system is the larynx and tracheal passage irritations. Tobacco smoke may cause swelling in air passage that produce the mucus which can block air ways. This cause lung infection and may damage the alveoli

SUMMARY

- Breathing consist of inhalation and exhalation. Inhalation is the entry of air in lungs and exhalation is the removal of air from lungs.
- Air passage is nasal cavity → pharynx → trachea → primary bronchi → secondary bronchi → tertiary bronchi → bronchiole → alveoli.
- Barrier between air and blood is just 1μ in thickness.
- Exchange of gasses from air to blood cell at alveoli is due to moist and large surface area containing membrane.
- Lungs are pair left lung is 2 lobed i.e., superior and inferior lobs. While right lung is 3 lobed i.e. superior, medial and inferior.
- Sound producing organs i.e. vocal cord present in larynx.
- Inter coastal muscles, ribs, diaphragm help in breathing.
- A membrane also protect the lungs called pleural membrane, which secrete slippery secretin which makes the max: expansion and contraction to lungs.
- One complete inhalation and exhalation is called breath cycle breathing rate depends upon breathing cycles takes place in unit time. i.e. 15-18 time/ minute in adult and healthy person.
- Lung capacity is the amount of air filled in lungs i.e. 5 liter in dermal breathing.
- Tidal volume is the amount of air required for survival.
- Rate of breathing depends upon the concentration of CO_2 and H^+ present in blood which are monitored by specific chemoreceptors present in blood vessels called aortic and carotid bodies.
- 97% O_2 is transported in body by R.B.C and the remaining 3% is transported by blood plasma.
- O_2 molecule of air are attached with hemoglobins to form oxyhemoglobing.
- CO_2 as by product inhale from body by three different ways

$$CO_2 + H_2O \quad H_2CO_3 \quad HCO^-_3 + H^+$$

$$H_2CO_3 + K^+ \quad KCOH_3$$

$$H_2CO_3 + Na^+ \quad NaHCO_3$$

- Pneumonia, pulmonary tuberculosis and emphysema are disease of lower respiratory disorder.

EXERCISE

1. **Encircle the correct choice**

(i) In which part of the respiratory system, gaseous exchange takes place?

(a) Alveoli (b) Pharynx
(c) Larynx (d) Trachea

(ii) Which of the following statements is true about involuntary breathing?

(a) It is controlled by the bronchioles
(b) It is controlled by the pulmonary arterioles
(c) It is controlled by the alveolar-capillary network
(d) It is controlled by the neurons, located in the medulla and pons

(iii) The tiny air sacs present in human lungs is called.

(a) Alveoli (b) Bronchus
(c) Bronchioles (d) All of the above

(iv) The exchange of gases between the external environment and the lungs.

(a) Respiration (b) External respiration
(c) Cellular respiration (d) None of the above

(v) The maximum volume of air contained in the lung by a full forced inhalation is called.

(a) Tidal volume (b) Vital capacity
(c) Ventilation rate (d) Total lung capacity

(vi) Which one of the following is correct regarding larynx?

(a) It houses the vocal cords
(b) It prevents the invading pathogens into the trachea
(c) It is an organ made of cartilage and connects the pharynx to the trachea
(d) All of the above.

(vii) Which of the following is the function of the trachea?

(a) Gaseous Exchange
(b) Filters the air we breathe
(c) Exhales the air from the body
(d) All of the above

(viii) Which of the following organs functions as an air conditioner?

(a) Larynx (b) Pharynx
(c) Nasal chambers (d) All of the above

(ix) Which of the following statements is true about the entry of air into the lungs?
- (a) Air enters the body and travels to the lungs through the mouth and the nose
- (b) Air enters the body and travels to the lungs through the esophagus and gullet
- (c) Air enters the body and travels to the lungs through the windpipe and the pores
- (d) Air enters the body and travels to the lungs through the nose and the nervous system.

2. Write short answers of the following questions:

1. When breathing process develop in animals?
2. What type of ventilation occur in human?
3. Why rate of breathing increases in human beings?
4. Why breathing of human being called negative pressure breathing?
5. How CO_2 from cell transport to lungs?
6. Which of the pigment is called respiratory pigment?
7. Draw a flow chart for the passage of air from external nares to alveoli.
8. Why hair and mucus glands are present in nostrils and trachea?
9. How exchange of gases occurs at the alveolar level?

3. Write detailed answers of the following questions:

1. What is respiration? Describe human respiratory system?
2. What are the 5 main functions of the respiratory system?
3. Draw labelled diagram of human respiratory system.
4. Describe the three regions of the pharynx and their functions.
5. What is the mechanism of inspiration and expiration?
6. Describe three ways in which carbon dioxide can be transported.

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